

Test 1A Review Questions

Student edition

These are questions worth focusing on for the test. Use them to guide your review of Modules 1A and 2A, the assigned videos, and the Retro and Cosmic Redshift slide quizzes. Brief answers are enough for most prompts; practice explaining the ideas in your own words. Use the clickable links below each question to find the relevant material, then write notes in the space provided.

The Universe and Cosmic Beginnings

1. What is the Universe, and what does cosmology study?

Find answers: [1AE1](#) | [Module 1A](#)

2. What is meant by a multiverse, and how does that idea relate to physical constants that permit life?

Find answers: [1AE1](#)

3. Approximately how old is the Universe, and how does its expansion help us reconstruct its past?

Find answers: [1AE1](#)

4. How does the raisin-cake analogy explain expansion without a special center or surrounding empty space?

Find answers: [1AE1](#)

The Early Universe and the Cosmic Microwave Background

5. Which light elements formed during primordial nucleosynthesis in the first 20 minutes?

Find answers: [Module 1A](#)

6. When was the cosmic microwave background released, and what change in matter allowed light to travel freely?

Find answers: [1AE1](#)

7. Who discovered the cosmic microwave background, and why was the discovery unexpected?

Find answers: [1AE1](#)

8. What is the CMB's temperature today, and why has it cooled since its release?

Find answers: [1AE1](#)

9. How does inflation help explain the CMB's nearly uniform temperature and the seeds of galaxies?

Find answers: [1AE1](#)

Dark Matter and Dark Energy

10. Approximately what percentages of the Universe are ordinary matter, dark matter, and dark energy, according to 1AE2?

Find answers: [1AE2](#)

11. How do galaxy rotation measurements provide evidence for dark matter?

Find answers: [1AE2](#)

12. What did observations of Type Ia supernovae reveal about cosmic expansion?

Find answers: [1AE2](#)

13. What is dark energy, and what is the Big Chill scenario for the Universe's future?

Find answers: [1AE2](#)

Olbers and Cosmic Redshift

14. What is Olbers' paradox, and which assumptions about the Universe produce it?

Find answers: [Olbers](#)

15. How do the Universe's finite age and the redshifting of light help explain the dark night sky?

Find answers: [Olbers](#) | [Cosmic Redshift](#)

16. How can light from a distant star reach Earth billions of years after the star has died?

Find answers: [Cosmic Redshift](#)

Aristotle and the Experimental Method

17. What did Aristotle mean by "Knowledge is the product of experience," and how does that connect to scientific evidence?

Find answers: [Aristotle](#) | [2AE1](#)

18. Why is the traditional list of five senses incomplete? Give two additional senses discussed in the videos.

Find answers: [Aristotle](#) | [2AE1](#)

19. How do telescopes and other instruments extend our senses beyond visible light?

Find answers: [2AE1](#)

20. What is the definition of an experiment? Why is it crucial for an experiment to be repeatable?

Find answers: [2AE1](#)

21. What is a scientific hypothesis?

Find answers: [2AE1](#)

22. How does the scientific meaning of “theory” differ from its everyday use to mean a guess?

Find answers: [2AE1](#)

Physical Laws and Conservation

23. How does a physical law differ from an advisory principle in the scientific method?

Find answers: [2AE2](#)

24. What is a conserved physical quantity? Give three examples from classical physics.

Find answers: [2AE1](#) | [Module 2A](#)

25. What does Galileo’s law of falling bodies predict for two objects of different masses dropped from the same height when air resistance is negligible?

Find answers: [2AE1](#)

26. What two objects did Commander Scott drop on the Moon, and how did the demonstration illustrate Galileo’s law?

Find answers: [Module 2A](#) | [2AE1](#)

Occam's Principle and the Principle of Falsifiability

27. When two competing hypotheses fit the same observations, what does Occam's principle advise, and does it guarantee that the simpler hypothesis is correct?

Find answers: [2AE2](#)

28. What does it mean for a hypothesis to be falsifiable?

Find answers: [2AE2](#) | [Aristotle](#)

Solar System Models and Retrograde Motion

29. What is apparent retrograde motion against the background stars? Does the planet actually reverse its orbit?

Find answers: [Retro](#)

30. Why does Mars appear to move backward when Earth overtakes it, and how is this related to opposition?

Find answers: [Retro](#)

31. Why does Venus exhibit retrograde motion near inferior conjunction, and which planet overtakes the other?

Find answers: [Retro](#)

32. How does the heliocentric explanation of retrograde motion illustrate Occam's principle?

Find answers: [Retro](#) | [2AE2](#)

The Zodiac and Cosmic Geometry

33. What is the zodiac, and how many constellations does the Sun actually pass through?

Find answers: [2AE2](#)

34. What do CMB measurements indicate about the Universe's geometry, and why does near-flatness alone not settle whether its total extent is infinite?

Find answers: [1AE1](#) | [Module 1A](#)

Cosmic Discoveries and Evidence

35. What contributions did Friedmann and Lemaître make to the idea of an expanding Universe?

Find answers: [1AE1](#)

36. What contributions did Hubble, Slipher, and Humason make to the idea of an expanding Universe?

Find answers: [1AE1](#)

37. Which two scientists are associated with the discovery of dark matter, and what observations did they use?

Find answers: [1AE2](#)

38. Is cosmic inflation a confirmed fact or a hypothesis in the episode? What distinction should you make between evidence for it and certainty?

Find answers: [1AE1](#)

39. Before the 1998 supernova results, why did astronomers expect cosmic expansion to slow down?

Find answers: [1AE2](#)
